SHREYA AMBASTHA

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SUMMARY

Results-driven CS student specialising in full-stack application engineering. Proficient in Java, Python, JavaScript, and modern frameworks including React, Node.js and Docker containerization. Seeking Software Engineer roles to design and build scalable, AI-powered solutions; keen to skill-up.

EDUCATION

Presidency University Bangalore — Bengaluru ,India
B.Tech in Computer Science & Engineering – CGPA: 8.66 — Jul 2023 – Jul 2027

Kairali School, Ranchi — Ranchi,India
Class 12th — Mar 2022

EXPERIENCE

Technical Intern – Web Development, Drobozone Innovative Centre Pvt. Ltd. — 16 Jul - 16 Aug 2025

- Contributed to full-stack web development tasks, building and maintaining responsive web pages and features as part of an Agile, sprint-driven technical team, contributing to innovative technology solutions for global business and technology teams.
- Assisted with debugging, testing, and refining application modules, following structured development workflows to ensure functional, reliable deliverables and smooth technical operations, with exposure to configuration management, testing and deployment.
- Collaborated cross-functionally on process improvements and day-to-day technical operations, gaining hands-on exposure to real-world software engineering practices, demonstrating strong communication, collaboration, ownership and accountability.
- Worked closely with senior developers to translate feature requirements into clean, reusable UI components, improving page load consistency across modules, while continuously learning new and emerging technologies to meet customer requirements.

TECHNICAL PROJECTS

CareBridge – Clinical Pathway Analytics Platform *Python · FastAPI · Next.js · TypeScript · Scikit-learn · FAISS · RAG · SQLite*

- Built an AI healthcare platform that extracts medical parameters from patient reports, runs risk analysis, and generates personalized clinical pathways, applying AI-powered, intelligent-system design to improve healthcare outcomes.
- Added a RAG-based AI assistant, specialist recommendations, and an OpenStreetMap doctor finder, with a customer-centric approach.
- Developed a Big Data pipeline to analyze pathway utilization, detect workflow deviations and bottlenecks, and visualize insights on an interactive dashboard, supporting data-driven business decisions.
- Engineered a FastAPI backend and a Next.js/TypeScript frontend to deliver a responsive, real-time clinical dashboard for care teams.
- Trained Scikit-learn risk-prediction models on extracted patient parameters to flag high-risk cases early and support triage decisions.
- Built a FAISS vector index over medical literature to power semantic retrieval for the RAG assistant, improving response relevance, with SQLite for structured patient and pathway data, following sound software architecture and design principles.

Big Data Analytics Platform *Python · Apache Spark · Hadoop HDFS · Kafka · Hive · SQL · Docker · Matplotlib*

- Built an end-to-end Big Data analytics platform integrating Apache Spark, Hadoop HDFS, and Hive for distributed data processing, storage, and SQL-based analytics, containerized with Docker for consistent, scalable deployment.
- Developed Spark ETL pipelines to transform and aggregate sales datasets, enabling scalable batch-processing workflows on Hadoop infrastructure.
- Integrated Apache Kafka for real-time data ingestion and streaming analytics using a producer-consumer, microservices-style architecture.
- Designed Hive tables and analytical queries, and created Matplotlib dashboards to visualize business insights and regional sales trends.
- Managed distributed storage on Hadoop HDFS, loading processed Spark outputs into Hive-backed tables to support downstream SQL analytics and reporting, aligned with functional and system requirements.

Task Management Web Application *React · Node.js · Express.js · MongoDB · JWT · Tailwind CSS*

- Built a full-stack task management app with a React frontend and a Node.js/Express REST API backend.
- Implemented JWT-based user authentication for secure login, registration, and logout, following secure, responsible development practices.
- Used MongoDB to store tasks, letting users create, update, and delete their tasks with a simple, responsive UI.
- Styled a responsive, mobile-friendly interface with Tailwind CSS, building reusable components for task lists, filters, and status boards.
- Built RESTful API endpoints in Express.js with middleware for request validation, error handling, and protected routes.
- Designed Mongoose schemas for users and tasks, enabling efficient querying and filtering by status, priority, and due date; tested modules.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#, SQL, Object-Oriented Programming (OOP), HTML, CSS

Frontend & Backend: React, Node.js, Express.js, FastAPI, REST APIs, .NET, HTML/CSS/JavaScript Front-End Styling, Open-Source Frameworks

Architecture & Practices: RESTful API Design, Testing, Agile, Requirements Analysis, Documentation, Application Performance, Security & Scalability, SDLC

AI & Data: LLMs, RAG, YOLO, NER, Scikit-learn, MongoDB, Redis, NoSQL Databases, MySQL

Concepts: Data Structures & Algorithms, Problem Solving, Microsoft Technologies

LEADERSHIP

- Core Member, Built Club – organized inter-college hackathons in Bangalore, demonstrating leadership, initiative and teamwork.
- Event Coordinator, TEDx Presidency University, showing ownership, adaptability and collaboration.
- Host Speaker, Decoding Boardroom Dynamics 2.0 – National Boardroom Simulation & NCLT Trial Competition.

CERTIFICATIONS & ACHIEVEMENTS

- Advanced Software Engineering Job Simulation (Walmart Global Tech – Forage) — Jun 2025.
- Software Engineering Job Simulation (JPMorgan Chase & Co. – Forage) — Jan 2026.
- JavaScript (Intermediate) Certification (HackerRank).
- AI Skills Passport Certification (Microsoft & EY).
- Research Paper: CareBridge – Clinical Pathway Analytics Platform – submitted to the 16th International Conference (Springer); under review.